

FORM 2
THE PATENT ACT 1970
(39 of 1970)
&
The Patents Rules, 2003
COMPLETE SPECIFICATION
(See section 10 and M.13)
LEAK PROOF MIXER JAR

Applicants(S) Name : New Horizon College of Engineering,
New Horizon Park,
Marathahalli outer ring road,
Bengaluru-560103.

Inventors(S) Name : Mrs. Priya N.
Senior Assistant Professor / Dept. of Information
Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,
Bengaluru-560103.

Mr. Poornachandra Tejaswi
Department of Information Science and Engineering,
New Horizon College of Engineering
New Horizon Knowledge Park
Outer Ring Road, Near Marathalli
Bellandur (P), Bangalore-560103

Mrs. J Karthiyayini
Senior Assistant Professor / Department of Information
Science and Engineering, New Horizon College of
Engineering
New Horizon Knowledge Park
Outer Ring Road, Near Marathalli,
Bellandur (P), Bangalore-560103

Mrs. Divya K. V.
Senior Assistant Professor / Department of Information
Science and Engineering,
New Horizon College of Engineering
New Horizon Knowledge Park
Outer Ring Road, Near Marathalli
Bellandur (P), Bangalore-560103

Dr. K Saravanan
Professor / Department of Information Science and
Engineering,
New Horizon College of Engineering
New Horizon Knowledge Park
Outer Ring Road, Near Marathalli
Bellandur (P), Bangalore-560103

The following specification particularly describes the invention and the manner in which it is to be performed

LEAK PROOF MIXER JAR

Title

Leak Proof Mixer Jar

Field of Invention

The present invention pertains to kitchen appliances, specifically to a mixer grinder jar lid that minimizes the risk of spillage during operation. The invention is particularly useful for high-speed mixing and grinding processes where splattering and overflow are common.

Background of the Invention

Mixer grinders are commonly used kitchen appliances for grinding, mixing, and blending food items. Traditional lids for these jars often fail to contain splashes and spills during high-speed operations, causing inconvenience and potential safety hazards. Spillage can lead to messy workspaces and wastage of ingredients. Moreover, liquid or semi-liquid substances can escape from the jar, posing an electrical hazard when they come in contact with the motor base. Thus, there is a need for an improved lid design that can prevent spillage while maintaining ease of use and compatibility with standard mixer grinder jars.

IN201921002350 - "MIXER GRINDER LID LOCK SAFETY PROTECTION DEVICE". The invention is related to safety locking mechanism, which is affixed in the jar. The mixer grinder works only if the jar lid and the jar is properly fix at their places. Fixing the jar lid into place allows the electrical conduction through a control switch. If the jar lid is removed the circuit is open and the mixer grinder cannot function.

IN202311028198 - A MIXER/ GRINDER JAR AND A METHOD OF PROTECTING THE LID FROM OPENING. A mixer/ grinder jar and a method of protecting the lid from opening. The present invention discloses a mixer/ grinder jar and a method of protecting the lid from opening. A mixer jar of present invention comprises: a pressure transducers, a means (CM) to detect the closeness of the lid of the jar; a means (CS) to generate the custom signal in the mixer by pressure transducers; a means (SR) to Set/ Reset the internal command if the lid is not closed tightly; and a downstream sensor to control turn ON or OFF of the mixer. The invention further discloses a method of protecting the lid from opening. A method for managing the turn on or off the mixer, wherein, the said method comprising: a step of detecting closeness of the lid of the jar; a step of passing the custom signal from the pressure transducers to the CS means; a step of giving internal command of Set/ Reset; and a step of activating downstream sensor to control turn ON/ OFF of the mixer.

Summary of the Invention

The present invention provides a mixer grinder jar lid (100) with an anti-spill mechanism that addresses the limitations of existing designs. The lid includes a robust locking system (102) that ensures a tight fit with the jar, preventing it from becoming loose or detached during operation. An innovative overflow channel (Figure 1, 101) is integrated along the inner circumference of the lid. This channel captures any liquid that rises during operation and redirects it back into the jar, preventing it from escaping through the edges. A flexible, food-grade silicone sealing ring is incorporated around the lid's periphery (103). The sealing ring conforms to the jar's opening, creating a leak-proof seal that adapts to variations in pressure caused by different speeds and food textures. The lid is equipped with a one-way ventilation valve (Figure 2, 101) that releases excess steam or pressure, reducing the chances of sudden spills or splatters. Strategically placed flaps (Figure 1, 101) within the lid's interior provide additional barriers to redirect splashes back into the jar, ensuring minimal wastage. This innovative design is easy to install, clean, and maintain, offering a practical solution to existing spillage issues faced by users of mixer grinders.

Detailed Description of the Invention

The following detailed description, along with the accompanying drawings, provides a comprehensive understanding of the invention's structure and function. The mixer grinder jar lid (100) is made of durable, heat-resistant plastic with a transparent top section for easy monitoring of the contents. The lid's inner surface is directed to splash back towards the center of the jar. The locking mechanism (102) consists of four interlocking components. A series of latches on the lid's body engage with corresponding edges on the jar's body, providing a firm hold. The locking system can be engaged or disengaged with a simple lift and latch action. The overflow channel is moulded into the lid's inner circumference (Figure 1, 101), with a sloped surface that guides liquid into the jar lid and store it. This channel is particularly effective for containing foamy or high-viscosity mixtures. The sealing ring is attached to the lid's lower edge (103), ensuring a snug fit. It can withstand temperature variations and remains flexible even after prolonged use. The ring can be removed for cleaning and reattached easily. The one-way ventilation valve (Figure 2, 101) is positioned at the lid's top center, allowing steam to escape without letting external air in. This valve helps maintain a stable pressure inside the jar, reducing the risk of spillage. These flaps (Figure 1, 101) are positioned in a radial pattern on the inner surface of the lid. They act as barriers that redirect splashes, minimizing leakage during vigorous mixing or grinding.

Brief Description of the Drawings

The mixer grinder jar lid (100) with anti-spill protection mechanism. The drawing shows the circular structure of the lid, with four rectangular locking clips (102) equidistantly spaced along the outer circumference. The central circular portion represents the lid's inner section, which fits securely onto the jar's opening. The circular ring indicates the sealing area with sloped surface that guides liquid into the jar lid that ensures a leak-proof fit (Figure 1, 101). The mixer grinder jar lid (100), illustrating the three-dimensional shape and positioning of the components. The four rectangular locking clips (102) are visible, extending outwards from the main circular body of the lid. The lid's surface is slightly raised in the center, where the sealing and locking mechanisms are integrated. The concentric circles represent different levels or sections of the lid, indicating the overflow prevention channels and one-way ventilation valve (Figure 2, 101).

Advantages of the Invention

The proposed mixer grinder jar lid with anti-spill protection offers the following advantages:

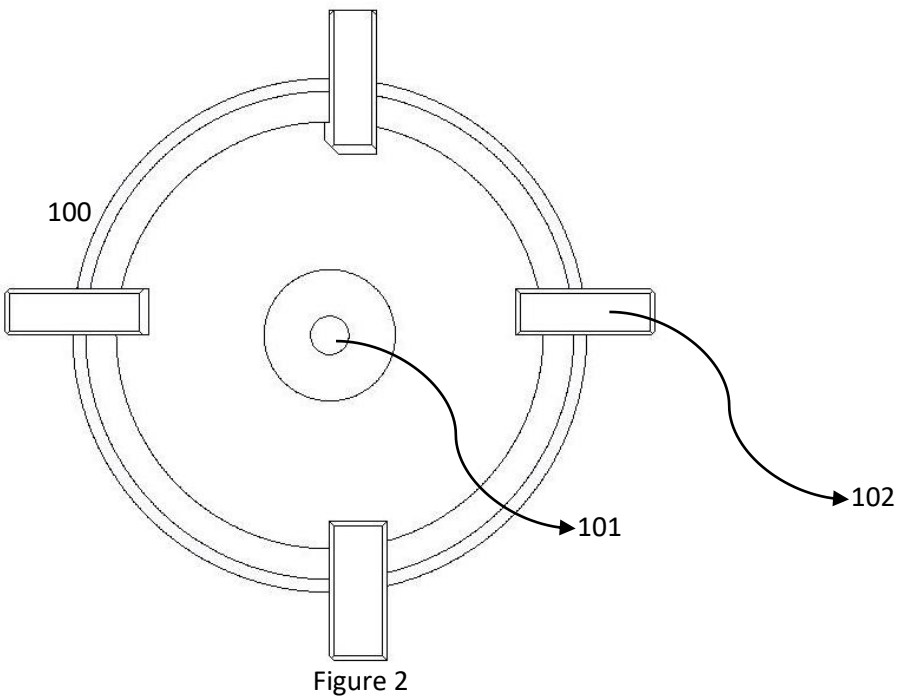
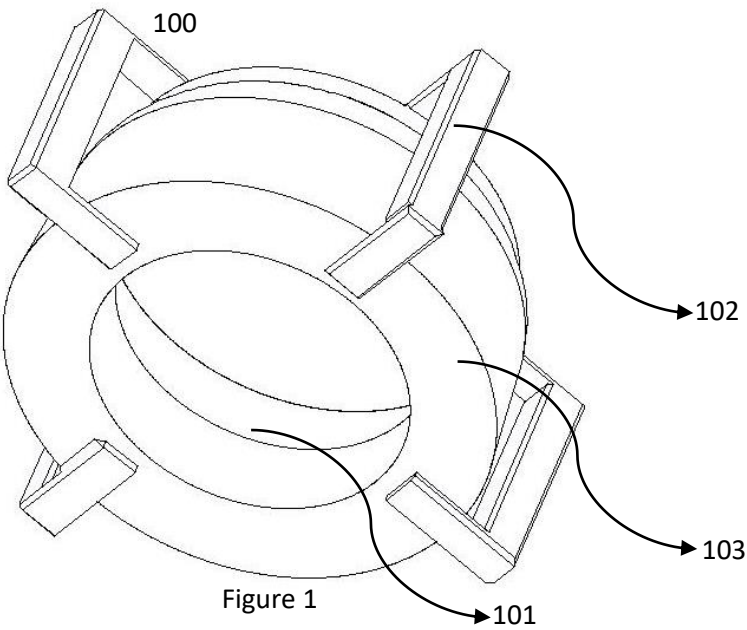
1. Prevents spills and splashes during high-speed operations.
2. Maintains a cleaner kitchen environment by reducing leakage.
3. Enhances safety by minimizing the risk of liquid reaching the electrical components.
4. Easy to install, clean, and compatible with various jar designs.
5. Provides a robust, durable solution that withstands temperature and pressure variations.

We Claim

1. A mixer grinder jar lid (100) with anti-spill protection, comprising a secure locking mechanism (102), an overflow prevention channel (Figure 1, 101), and a flexible sealing ring (103) that provides a leak-proof seal between the lid and the jar.
2. A mixer grinder jar lid (100) with anti-spill protection of claim 1, wherein the overflow prevention channel (Figure 1, 101) is integrated along the inner circumference of the lid and configured to redirect liquid into the jar lid.
3. A mixer grinder jar lid (100) with anti-spill protection of claim 1, further comprising a one-way ventilation valve (103) positioned at the top center of the lid, designed to release excess steam or pressure.
4. A mixer grinder jar lid (100) with anti-spill protection of claim 1, further including spill guard flaps (Figure 1, 101) arranged on the inner surface of the lid to reduce splatter and contain splashes within the lid.
5. A mixer grinder jar lid (100) with anti-spill protection of claim 1, wherein the flexible sealing ring (103) is made of food-grade silicone and is removably attached to the lid's lower edge.
6. A mixer grinder jar lid (100) with anti-spill protection of claim 1, wherein the secure locking mechanism (102) comprises interlocking grooves and projections that provide a stable fit and prevent detachment during operation.

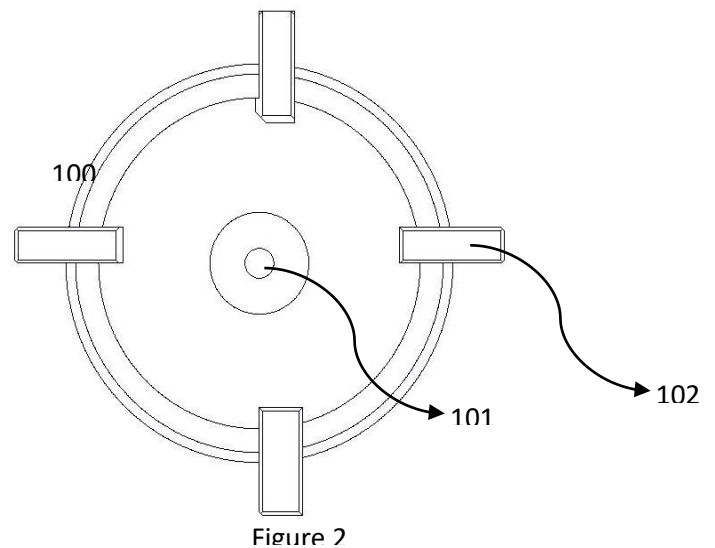
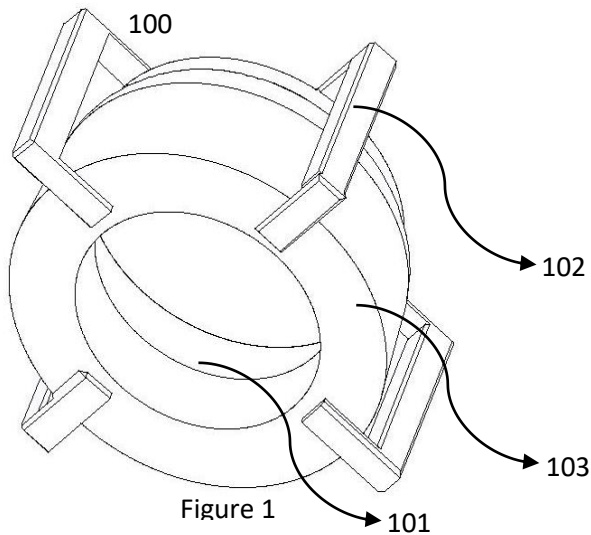
Date: 04/10/2024

**Dr. Manjunatha,
Principal, New Horizon College of Engineering**



Abstract

The mixer grinder jar lid (100) with anti-spill protection invention relates to an improved lid for any mixer grinder jar, designed to prevent spillage during high-speed operations. The lid features an innovative anti-spill mechanism comprising a secure locking system (102), an overflow prevention channel (Figure 1, 101), and a flexible sealing ring (103). The mechanism ensures a leak-proof seal and prevents accidental spillage during mixing or grinding, thereby maintaining cleanliness and reducing waste. This design can be applied to various types of mixer jars and is easy to manufacture and install. The mixer grinder jar lid (100) with anti-spill protection described in this patent provides a significant improvement over conventional lids by integrating features that prevent spillage, enhance user safety, and maintain cleanliness. The design is versatile and adaptable to a wide range of mixer grinder models, making it an ideal addition to modern kitchen appliances.



Date: 04/10/2024

Dr. Manjunatha,
Principal, New Horizon College of Engineering